Gaussian distribution, we can determine whether a variable (someone's house) is similar enough to the variable we're interested in (our own house) to influence our posterior belief and to what extent.

2.2 Modeling correlations with multivariate Gaussian distributions and Bayesian updates

In this section, we will learn about multivariate Gaussian distributions (or multivariate Gaussians, or simply Gaussians) and see how they facilitate the update rule we saw above. This will serve as the basis for our subsequent discussion on Gaussian processes.

2.2.1 Using multivariate Gaussian distributions to jointly model multiple variables

We will first learn about what multivariate Gaussians are and what they can model. We will see that with a covariance matrix, a multivariate Gaussian describes not only the behavior of the individual random variables, but also the correlation of these variables.

First, let's consider normal distributions, a.k.a. the bell curve. Normal distributions are highly common in the real world and are used to model a wide range of quantities; examples include height, IQ, income, and weight at birth.

A multivariate Gaussian distribution is what we would use when there is more than one quantity that we want to model. To do this, we aggregate these quantities into a vector of random variables, and this vector is then said to follow a multivariate Gaussian distribution. This aggregation is depicted in Figure 2.4.

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